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Remaining useful life prediction of turbofan engines based on dual attention mechanism guided parallel CNN-LSTM

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Abstract

Remaining useful life (RUL) prediction is an valuable research task for predictive health management of aero engines, which is crucial to enhance the safety, dependability and economy of the motor. Accurate prediction of remaining engine service life is an important means for the effectively monitoring of engine operating conditions. The forecasting accuracy of turbofan RUL is inadequate with the traditional single-parameter, single-stage mode. To boost the precision of aero-engine RUL anticipation, a novel mode is recommended, based on a parallel convolutional neural network (CNN) with a long and short-term memory (LSTM) neural network and a dual attention mechanism, named PCLD. The degradation information directly from time series sensor data. The advantages of CNN networks and LSTM networks in feature mining and time series processing, respectively, are employed to process time series data, which is conducive to preventing the loss of important element in the data. At last, the experimental results on the aero-engine performance recession dataset C-MAPSS demonstrate that the method outperforms the currently popular models with better robustness and higher prediction accuracy.

Keywords: Remaining useful life prediction, parallel CNN-LSTM, dual attention mechanism, aero-engine

1. Introduction

Remaining useful life (RUL) forecast of machinery and equipment is of great interest to academia and industry, but also as an important component of prognostics and health management. [1–4]. Making accurate RUL predictions is beneficial to preventing major equipment accidents, developing reasonable maintenance programs, ensuring working conditions, and

preventing unexpected accidents [5–7]. As one of the important components of the aircraft power system, the reliable and efficient operation of the aero-engine is crucial to the flight safety of the aircraft. Therefore real-time, accurate prediction of aero-engine RUL and early decision making can significantly reduce the probability of disaster and carry out maintenance. There are two main types of RUL forecasting methods: model-based and data-driven [8–11]. The model-based approach utilizes the knowledge of the principles of system failure. The aim is to establishment of mathematical formulas for system degradation for RUL assessment. Model-based

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prediction is carried out on the basis of equipment degradation mechanisms, and the parameters of the model are selected in conjunction with monitoring data [12, 13]. It has a high prediction accuracy provided it is accurately modeled, but with the complexity of the device structure, accurate modelling is often not possible [14, 15]. As a result, model-driven approaches are more expensive to develop and less generalizable. In addition, the data-driven approach is based on state prediction based on monitored past data without involving modelling knowledge, which reduces errors due to knowledge limitations in the prediction process, so it is more in line with the requirements of life prediction nowadays, and has become the mainstream approach nowadays.

Data-driven RUL prediction based methods are more widely used in the field of turbofan engine RUL prediction. Shallow machine learning (SML) [16–18], deep machine learning [19, 20], etc are widely used for turbofan engine RUL anticipate. artificial neural networks [21], extreme learning machines [18], Random Forests [17], Support Vector Machines (SVM) [16], etc which are included in SML, can be used to implement RUL predictions. For example, Khelif *et al* completed a hybrid particle swarm parameter optimisation model based on SVM applied to RUL prediction for turbofan engines [22]. García Nieto *et al* used support vector regression to fit the RUL [23]. As mentioned above, RUL prediction first requires processing of the extracted monitored multi-performance parameter data. However, due to the small number of hidden layers within the SML model and the simple internal structure, the extracted features are usually shallow representations of the data information. The RUL prediction generalization ability of the SML approach will be limited when dealing with complex data. So the RUL prediction model has to go from shallow to deep in the face of the aero-engine with multi-sensory data. Improvements in computer processing power and the availability of efficient optimization algorithms provide more favourable tools for residual life prediction. The deep learning process is mainly a superposition of multiple layers of shallow learning, performing layer-by-layer feature transformations, transforming the original data from the feature shape of the original space to a more advanced feature space, solving the complex problem of high-dimensional features, and ensuring that the prediction task is carried out smoothly. Among the most representative deep learning algorithms are convolutional neural networks (CNN) [24] and long short-term memory neural networks (LSTM) [25]. Yu *et al* implemented a new BLSTM model to improve the RUL prediction accuracy [26]. Babu *et al* attempted to use CNN for predictive estimation of RUL, using a sliding time window for feature enhancement of the original data, which allowed the CNN to more adequately perform feature learning for extraction [27]. Since whole-life data contains information not only time series information but also feature information. So hybrid models combining basic deep learning models appear. Currently, multi-model hybrid deep learning is widely used, and hybrid models consisting of CNN models and LSTM models are also attracting a lot of attention. CNN networks can deeply mine the deep characterization of

data, and LSTM networks can deal with the temporal features of the data, so the accuracy of aero-engine prediction can be effectively improved by using hybrid models. Gao *et al* [28] proposed a one-dimensional CNN-BiLSTM hybrid neural network in the direction of lithium-ion battery residual life prediction to mine historical data for deep features and to preserve past information for highly accurate predictions. Li *et al* proposing a Convolutional Attention Block Optimisation CNN-LSTM Based Approach for RUL Prediction of Aero Engines [29].

Although in the serial CNN-LSTM model architecture, the output of the CNN is used as the input to the LSTM and the prediction results are accurate. However, extracting features in the CNN layer affects the training output of the LSTM and the ability to extract temporal features is not effectively utilized. A parallel CNN-LSTM model architecture is proposed. The problem of gradient vanishing will be especially prominent as the size of the dataset becomes larger and the length of the data sequence increases, attention mechanisms can be effective in solving this problem. Based on the high-dimensional performance parameter data obtained from multi-sensor monitoring of aero-engine, a parallel CNN-LSTM model structure guided by a dual-attention mechanism is proposed, which can fully exploit the deep features and utilise the temporal features to make up for the problems of information loss and gradient vanishing in the tandem model. The dual-attention mechanism directs the two channels, CNN and LSTM, and focuses on the important information by applying weights to the feature dimensions and time steps, respectively. This paper has the following contributions: (1) A parallel CNN-LSTM structure is proposed to allow the feature mining capability of CNN and the temporal processing capability of LSTM to be effectively utilised for accurate lifetime prediction. (2) A dual attention mechanism (DAM) module is introduced, which applies feature attention mechanism and temporal attention mechanism to CNN and LSTM paths, respectively, to focus on the important information in the temporal information. (3) Exponentially weighted shift smoothing processing is used to reduce the influence of noise on prediction.

The rest of this article is presented as follows: section 2 is an introduction to the workings of individual structure maps for LSTM networks. Section 3 presents a structural diagram of the proposed method and a description of the whole prediction process. Section 4 is a description of the data used for the experiments and the labelling of the data along with the series of experiments conducted on this dataset. Section 5 conducting thesis summaries.

2. Methodologies

This paper focuses on RUL prediction for turbofan engines. In this subsection, in connection with the problems of RUL anticipation, this paper introduces parallel CNN-LSTM and dual-attention mechanism network models to improve capacity of the model to mine degraded information and achieve high-precision prediction. The dual-attention mechanism, on the

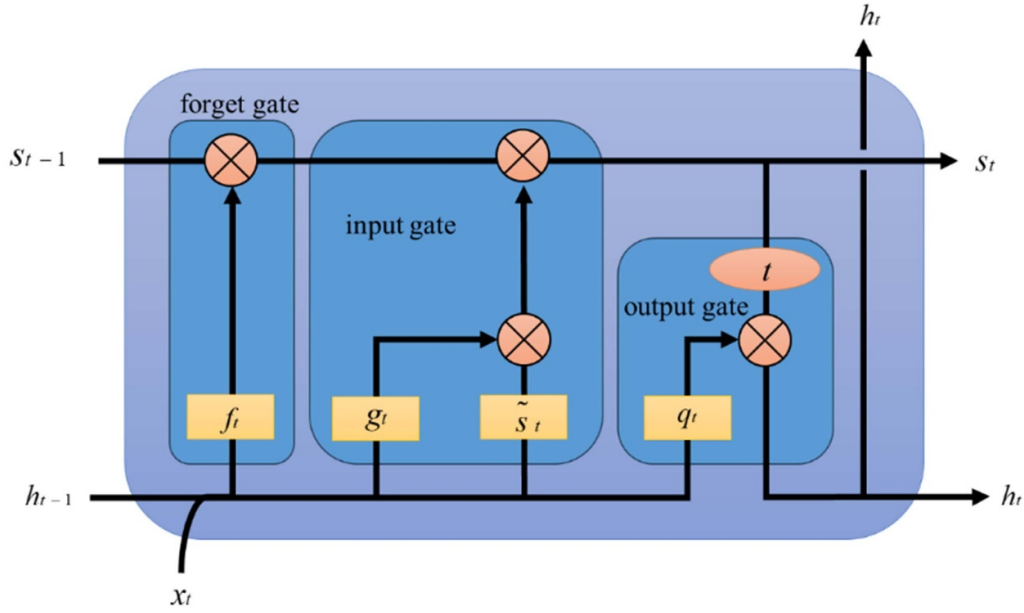


Figure 1. Diagram of LSTM network structure.

other hand, assigns weights at time steps and feature dimensions to highlight important feature information in degraded information. The model contains an important number of network modules, and the LSTM module is able to well deal with the temporal features in data, providing the necessary analyses for RUL prediction.

2.1. LSTM network

Hochreiter and Schmidhuber proposed a special class of recurrent neural networks (RNN) containing three gates in 1997, which was designed to solve the RNN gradient vanishing and explosion problem. LSTM can recognize and memorize long-term dependencies between data by controlling input gates, forgetting gates and output gates. As shown in figure 1 LSTM structure. The LSTM contains four interactive neural network layers. The forgetting gate f_t in LSTM is used to control how much of the output h_{t-1} passed from the previous moment needs to be forgotten, and the input gate g_t is used to control how much of the input state $\tilde{s}_t$ of the current moment needs to be saved to the current new state s_t , and finally the output gate q_t is utilized to pass the new state s_t of the current moment to the current output h_t .

Where x_t represents the current input, h_t is the current output, h_{t-1} is the output of the previous moment, s_t is the new state value, and s_{t-1} is the state value of the previous moment. Firstly, the output of the previous moment h_{t-1} and the input of the current moment x_t are used to calculate the forgetting gate f_t , input gate g_t , output gate q_t and input state $\tilde{s}_t$, and then the current state s_t is updated by combining the forgetting gate f_t and input gate g_t . The flow process within the block of the LSTM cell is computed by the formula:

$$f_t = \sigma(b_f + U_f x_t + W_f h_{t-1}) \quad (1)$$

$$g_t = \sigma(b_g + U_g x_t + W_g h_{t-1}) \quad (2)$$

$$\tilde{s}_t = \tanh(b + U x_t + W h_{t-1}) \quad (3)$$

$$s_t = f_t s_{t-1} + g_t \tilde{s}_t \quad (4)$$

$$q_t = \sigma(b_q + U_q + W_h h_{t-1}) \quad (5)$$

$$h_t = q_t \tanh(s_t) \quad (6)$$

where: $\sigma(\cdot)$ represents the sigmoid activation function; U_f, U_g, U, U_q are the input weights of the forgetting gate, the input gate, the unit state, and the output gate, respectively; W_f, W_g, W, W_q are the cyclic weights of each gating unit, respectively; and b_f, b_g, b, b_q are the biases of each gating unit, respectively.

3. The proposed method

3.1. Parallel CNN-LSTM and DAMs

In the face of the above challenges, the novel structure of parallel CNN-LSTM and DAM proposed in this paper predicts the aero engine health state. The full-life data of the turbofan engine is fed into CNN and LSTM, which not only has the time modelling capability of LSTM, but also can describe local features like CNN. This enables neural networks to capture temporal and spatial features, not just temporal features, while using attention mechanisms to focus on important features. Compared with the sequential model structure of CNN and LSTM, LSTM can be used more effectively.

The structural framework of the prediction method proposed in this article is shown in figure 2. In the structure of the RUL anticipate method proposed in this paper, a channel is a stack of two layers of LSTM, and the other channel is

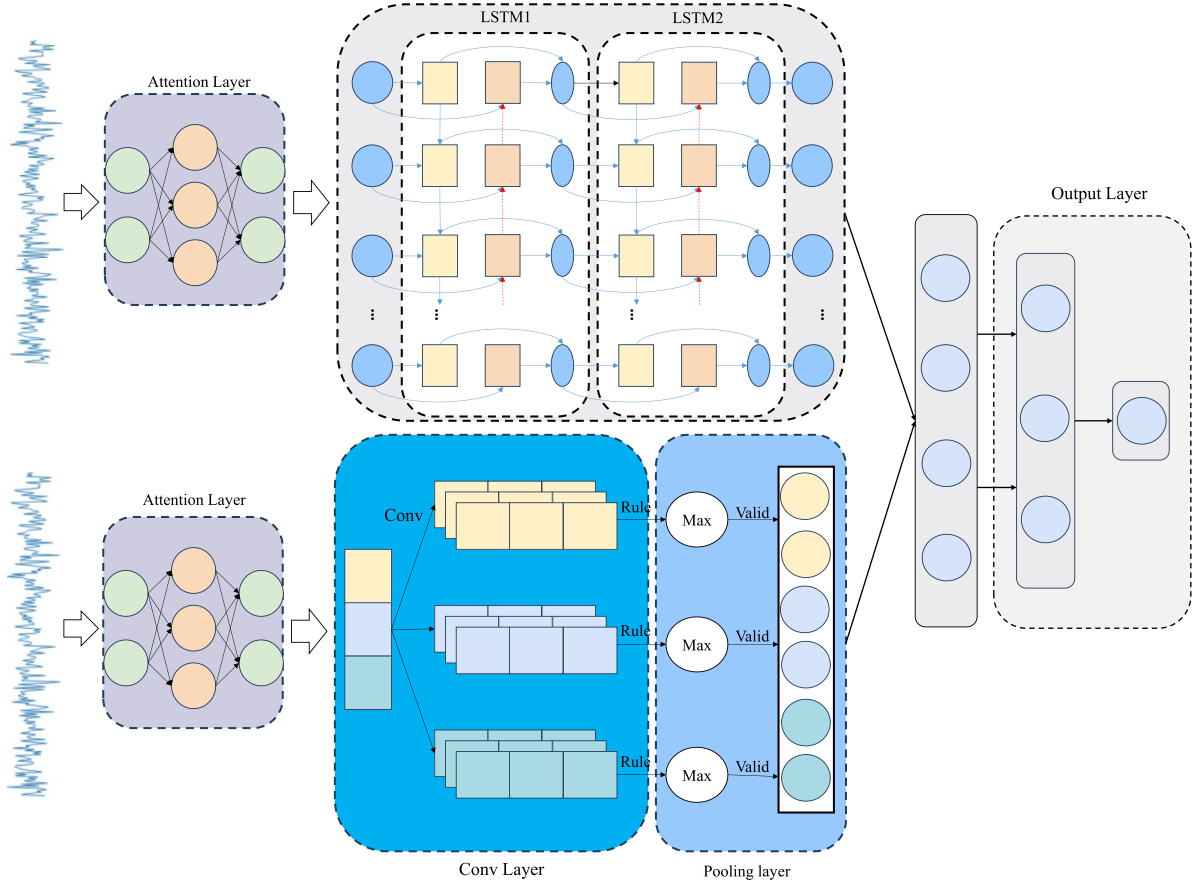


Figure 2. PCLD network framework.

stacked with two layers of CNN to extract mining depth features. The dual channel mechanism can capture richer information, thus improving the performance and robustness of the model. In other words, a single channel is a single convolutional layer to extract features and form feature mappings, while a double channel is a process of input data through two parallel convolutional layers at the same time, which can capture and combine features more comprehensively and richly. The detailed structure diagram is shown as follows: First, the multi-dimensional characteristic time series data of the engine is converted into multi-input array. The data are then input into the attention mechanism layer, and the proposed DAM is the contribution of different time steps and feature dimension to RUL prediction. After passing the attention mechanism layer, the data are input to the CNN and LSTM layers respectively for training. For highly accurate predictions, the following operations are carried out: The same fill and rectification linear unit activation function is used in the CNN layer; The LSTM layers all use Ridge regularization, calculating the sum of the squared values of the model coefficients to add to the loss function. Ridge regularization prevents overfitting by Splitting the impact of one feature to multiple features. Ridge regularization prevents overfitting by spreading the influence of a single feature across multiple features. Finally, Fusion of

features from dual-channel outputs for prediction as input to final prediction.

A DAM is invoked for the effects of time steps and features on prediction. The dual-attention mechanism is not only able to realize the weight allocation on time-specific step, but also on feature dimensions. The dual-attention mechanism plays an important role in each part of the input data by assigning weights to both parts. The attention mechanism is implemented as follows: (1) Input: the input sequence is $V = \{x_1, x_2 \dots x_m\}$. (2) Calculating Attention Weighting: Let $K_i = V$. The attention weights are calculated as follows:

$$\varphi_i = \text{softmax}(S) = \frac{e^{S(Q,K)}}{\sum_{i=1}^n e^{S(Q,K)}} \quad (7)$$

$$S(Q,K) = V^T \tanh(W_k K + W_q Q) \quad (8)$$

where Q is the lookup matrix and after mapping Q and K to different vector spaces, weights are calculated from similarity. Where Softmax can be normalized. (3) Weighted summation: $\text{Attention}(Q, \text{Source}) = \sum_{i=1}^n V \cdot \varphi_t$. The above process enables the allocation of attention matrices for feature dimensions and time steps.

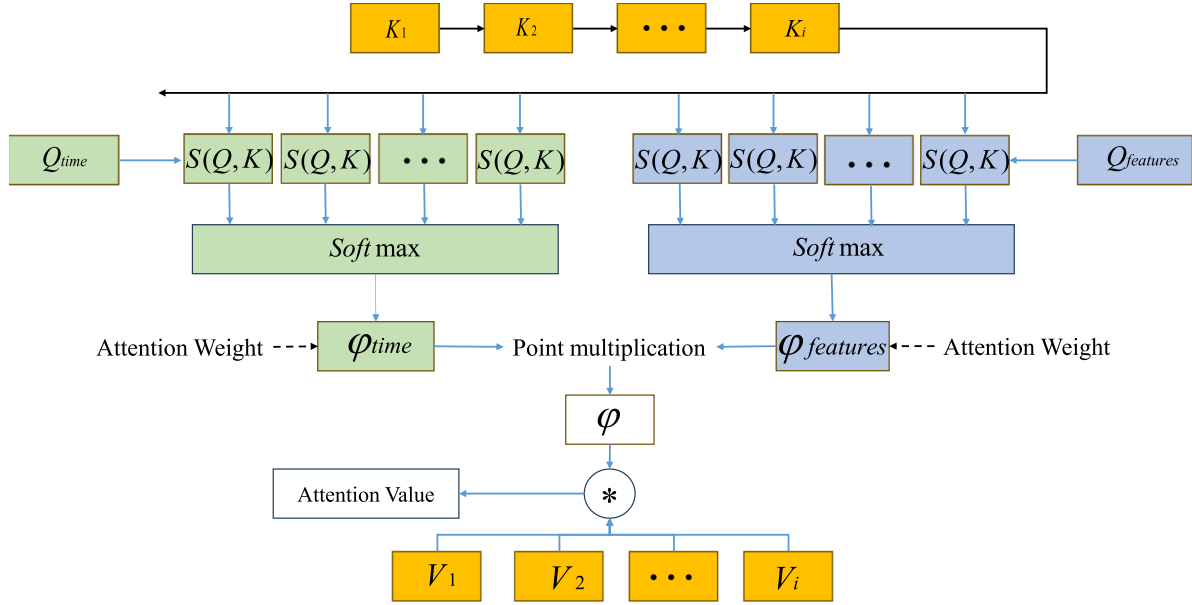


Figure 3. Methodological map of the dual-attention mechanism.

$$\varphi_{\text{timesteps}} : \begin{bmatrix} a_1 & \cdots & a_1 \\ \vdots & \ddots & \vdots \\ a_j & \cdots & a_j \end{bmatrix} \quad (9)$$

$$\varphi_{\text{feature}} : \begin{bmatrix} x_1 & \cdots & x_i \\ \vdots & \ddots & \vdots \\ x_1 & \cdots & x_i \end{bmatrix}. \quad (10)$$

Multiplying the two attentional weights yields

$$\varphi = \begin{bmatrix} a_1 \times x_1 & \cdots & a_1 \times x_i \\ \vdots & \ddots & \vdots \\ a_j \times x_1 & \cdots & a_j \times x_i \end{bmatrix}. \quad (11)$$

With equation (12) integrating the individual elements and weights of the inputs, and according to the dual-attention mechanism diagram shown in figure 3 shows that the weights can be assigned both on the feature dimensions and the time step without any influence on each other, and the validation of the validity is carried out afterwards.

3.2. RUL prediction framework

The projections framework of RUL is shown in figure 4, and the steps are as follows:

Step 1: Full-life engine degradation signals are captured using 21 sensors;

Step 2: By analysing the trends in the whole-life data and selecting the feature data that can be used, a maximum-minimum normalisation process is performed so that the data standards are in the same range and the use of weighted exponential moving averages for whole-life data with clear trends highlights the importance of the data at the current moment in time;

Step 3: A segmented linear degradation model with a threshold of 128 is set based on fluctuations in the whole-life data. Degradation labels set to full-life data;

Step 4: A moving time window is used in the training process of the neural network. The size of the time window is a two-dimensional matrix consisting of multidimensional feature parameters and time steps that are slid forward one time unit at a time along the time direction. The integrity of the input data is guaranteed;

Step 5: Input the extracted features into the PCLD model for RUL anticipation, and finally get the predicted RUL.

4. Data description and labelling

This section focuses on validating the feasibility of the proposed method using data published by NASA, and two subsets of the C-MAPSS dataset, FD001 and FD003, are used to validate the model here.

4.1. Description of data

The data are from NASA's publicly available aero-engine performance degradation simulation dataset, known as the C-MAPSS dataset, in which the aero-engine data are simulated for General Motors' GE90 series of large, high-containment-ratio turbofan engines. Its structure is schematically shown in figure 5 [30].

The main components of a turbofan engine are the fan (Fan), the compressor (PC) and the turbine (PT), with annular combustion chambers and nozzles consistent with those of the same series of engines. The engine's fan/pressor, turbine, combustion chamber, and nozzle are detected separately

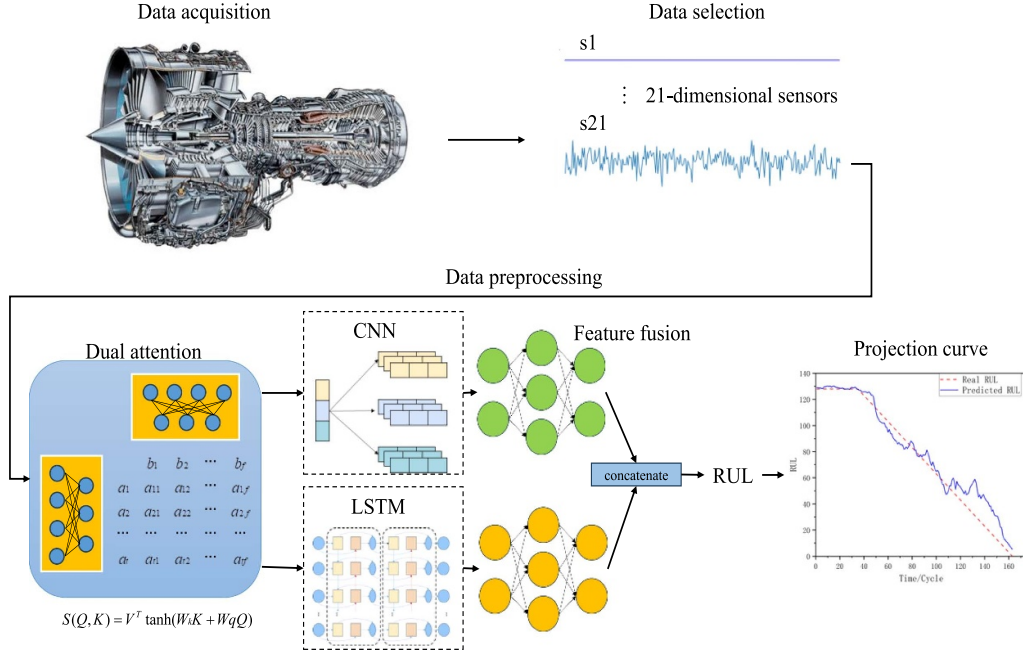


Figure 4. RUL Forecasting Process Framework.

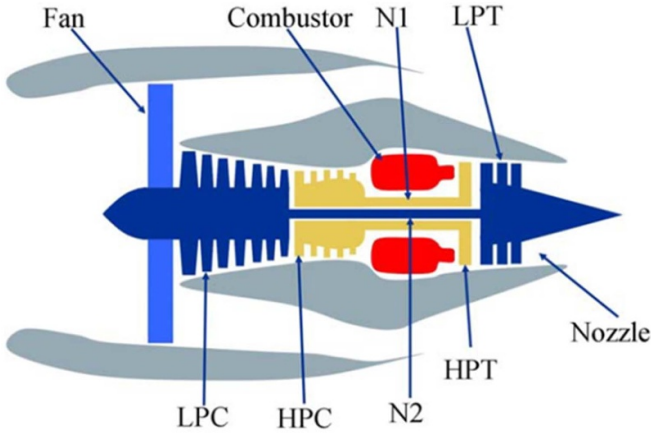


Figure 5. Schematic diagram of an aero-engine structure.

using different sensors. Table 1, the C-MAPSS dataset contains six main working conditions. In the engine data, six different operating conditions in the data set are realized by setting different flight altitude (H), Mach number (Ma) and temperature (T). Table 2, the C-MAPSS dataset comprises four data subsets such as FD001-FD004.

Each dataset is subdivided into a training set and a test machine, e.g., the training set of FD001 is train_FD001, and the test set is test_FD001. In this case, the training dataset contains full-life degradation data from health to failure for multiple engines, and, the test dataset records information on the flight cycles of multiple turbofan engines from health to before failure. Each data subset is a 26-dimensional time series of, from left to right, engine number, number of flight cycles, altitude, Mach number, temperature, and sensor measurements for 21

Table 1. Data set information.

Condition	Parameter		
	H	Ma	T
Condition1	0	0.00	100
Condition2	10	0.25	100
Condition3	20	0.70	100
Condition4	25	0.62	60
Condition5	35	0.84	100
Condition6	42	0.84	100

Table 2. Subset of data.

Data sets	Cycle	Condition	Faults
FD001	100	1	1
FD002	260	6	1
FD003	100	1	2
FD004	248	6	2

performance parameters. These 21 performance sensor data do not all describe the engine degradation trend, as shown in figure 6, where the full-life measurement curves for sensors 1, 5, 6, 10, 16, 18, and 19 are stable, and contain less information about the engine degradation, and so it does not apply to RUL forecasts. Finally the degradation data of other 14 performance parameters are input to the prediction model to predict them.

4.2. Data preprocessing

Data smoothing: Sensor-monitored aero-engine data with complex noise has a certain impact on the prediction results,

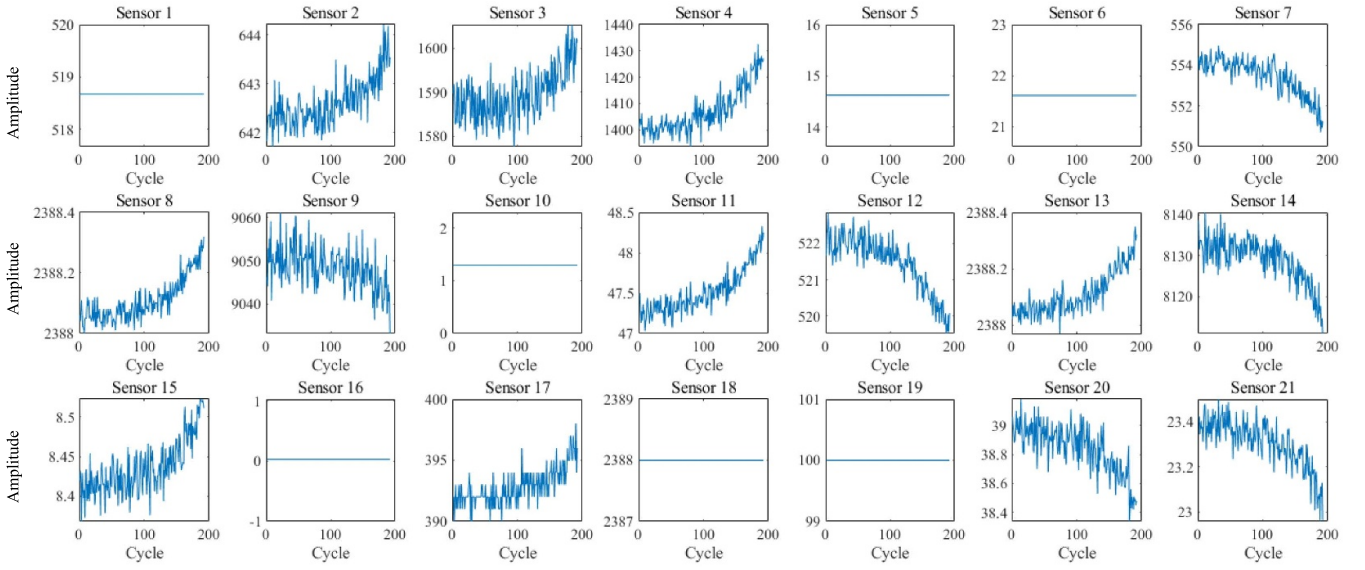


Figure 6. Sensor degradation data.

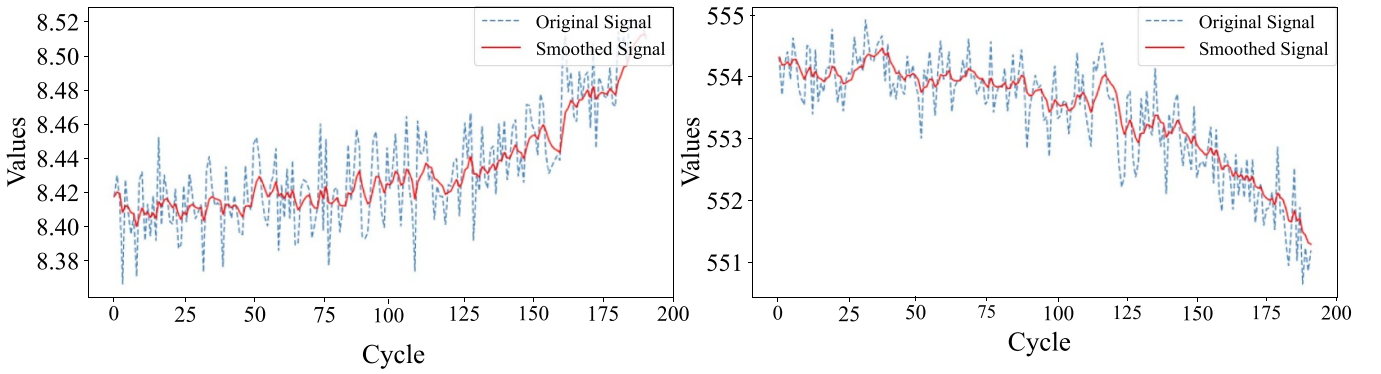


Figure 7. Data smoothing.

this paper uses a weighted moving average smoothing process to make the degradation trend more obvious and reduce noise interference with prediction results. The calculation formula is as follows:

$$M_{tw} = \frac{w_1 y_1 + w_2 y_2 + \dots + w_N y_{t-N+1}}{w_1 + w_2 + \dots + w_N}, t \geq N \quad (12)$$

where M_{tw} is the t -period weighted moving average: w_i is the weight of y_{t-i+1} , which reflects the importance of the corresponding y_t in the weighted average. Take a sensor as an example, as shown in figure 7 for a sensor data after smoothing, after smoothing the sensor data to eliminate some of the random fluctuations, its degradation process is more clear, monotonicity is more obvious. Weighted moving average method is to give different weights to the observed values, obtain the moving average according to different weights, and determine the predicted value based on the last moving average. For the observations close to the forecast period, a higher weight value is given, and for the observations far away from the forecast period, a smaller weight value is correspondingly

given. The effects of each observation value on the forecast value are adjusted with different weight values, therefore, the predicted value can better reflect the future development trend of the data. Then the data is split into time feature window data.

Data normalization: Since the sensors in the engine collect different parameters, the magnitudes of these parameters are not in the same range. In order to efficiently train the model and scale the sensor data to the range (0, 1), this paper uses min-max normalization to divide the difference between x and $x_{\min}$ and the difference between $x_{\max}$ and $x_{\min}$ to obtain the normalized value x' .

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (13)$$

Degradation Label Setting: Considering the important role of degraded data labelling in the prediction process, the RUL labelling needs to be added to the engine by including the cycles that the engine operates from the initial state to the occurrence of failures in the full life cycle data of the training

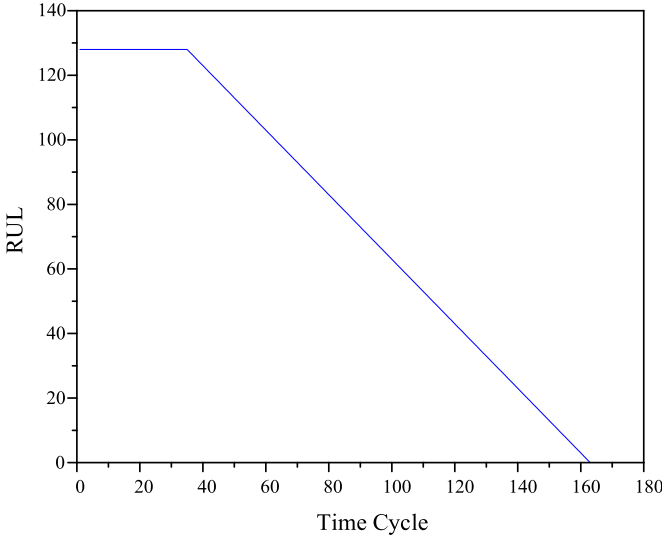


Figure 8. Segmented linear degeneration model.

set. In the initial stage of aero-engine operation, each component performs well, and wear and aging occurs during engine operation, resulting in a serious decline in the remaining life, a segmented linear degradation model is proposed to differentiate between the various stages of the engine. The whole life of an aero-engine is divided into a period of constancy and a period of degradation. Generally speaking, the threshold of the piecework linear degradation model for turbofan engine life prediction is set between 125 and 130. When we analyse the fluctuation of the original data, we find that when the RUL value drops to 128 time, the engine will enter the degradation stage. So the RUL value for the constant period is set to 128, as shown in figure 8, to demonstrate the degradation model curve.

Set up a sliding window of multidimensional time series, and their corresponding RUL labels, and use them as input for training. Window data $X \in R_{t \times n}$ is denoted as:

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_T \end{bmatrix} = \begin{bmatrix} X_{1,1} & \cdots & X_{1,n} \\ \vdots & \ddots & \vdots \\ X_{T,1} & \cdots & X_{T,n} \end{bmatrix} \quad (14)$$

where n is the sensor number, T is the window size, and X consists of three types of data, here $X = [X_{he}, X_{ts}, X_{dg}]$, denoting window data for healthy periods, data for excessive periods, and data for recessionary periods, respectively. In this study, a high-quality prediction model was trained to map the set time window X to the target remaining lifetime $yRUL$ of the next cycle. This problem can be expressed as:

$$\hat{y}RUL = \varphi(X) \quad (15)$$

where $yRUL$ is the actual RUL and $\hat{y}RUL$ is the predicted RUL. After data processing, the window data of length T is input to the deep learning model for prediction.

Degraded feature extraction: Firstly, considering the various monitoring data types of turbofan engines, we reduce the RUL prediction error by evaluating the monotony and robustness characteristics. For the eigenvector $F = [f(1), f(2), \dots, f(K)]$, time series $T = [t(1), t(2), \dots, t(K)]$, where $f(t_k)$ represents the eigenvalue obtained at time t , where $k = 1, 2, \dots, K$, K represents the length of time. Using the exponential weighted moving average method, the feature sequence is divided into two parts, one is the stationary trend term $f_T(t_k)$, and the other is the random margin term $f_R(t_k)$. $f(t_k) = f_T(t_k) + f_R(t_k)$. The calculation formula of the exponential weighted moving average method is as follows:

$$f_T(t_k) = \varepsilon f_T(t_{k-1}) + (1 - \varepsilon)f(t_k) \quad (16)$$

The correlation evaluation index between feature series F and time series T is denoted as $\text{Corr}(F, T)$, and the formula is as follows:

$$\text{Corr}(F, T) = \frac{|K \sum_k f_T(k) t_k - \sum_k f_T(k) \sum_k t_k|}{\sqrt{[\sum_k f_T(k)^2 - (\sum_k f_T(k))^2] [\sum_k t_k^2 - (\sum_k t_k)^2]}} \quad (17)$$

The point tonality evaluation index of F is denoted by the formula $\text{Mno}(F) \text{Mon}(F)$ as follows:

$$\text{Mon}(F) = \frac{1}{K-1} \left| \sum_k \delta(f_T(k+1) - f_T(k)) - \sum_k \delta(f_T(K) - f_T(k+1)) \right| \quad (18)$$

where, $\delta(t)$ is a unit step signal, which is specifically expressed as follows:

$$\delta(t) = \begin{cases} 1, t \geq 0 \\ 0, t < 0 \end{cases} \quad (19)$$

The robustness evaluation index of feature sequence is denoted as: $\text{Rob}(F)$, and the formula is as follows:

$$\text{Rob}(F) = \frac{1}{K} \sum_k \exp\left(-\left|\frac{f_R(k)}{f(k)}\right|\right) \quad (20)$$

The value range of the above three feature evaluation indicators is $[0, 1]$. The larger the value, the higher the feature score. Before calculating the three evaluation indicators, the input data need to be normalized. The feature sequence F is normalized to $[0, 1]$, and the time series T is normalized to $[-1, 0]$. -1 indicates the initial monitoring time, and 0 indicates the final

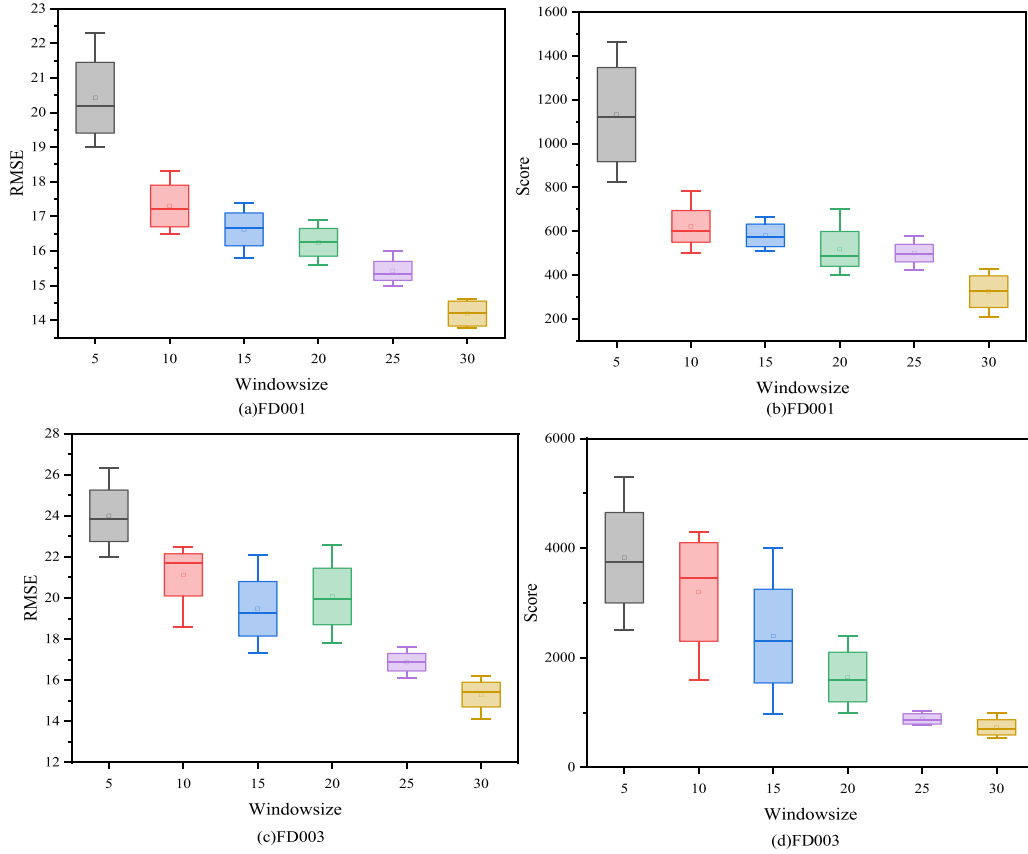


Figure 9. Predicted results under different windows.

failure time. Three evaluation indicators were used to evaluate the above 21 features and extract the features that meet the criteria from the data.

Evaluation metrics: Regarding the model proposed in this paper, 2 performance metrics are selected and defined as follows.

root mean square error (RMSE), The square root of the ratio of the square of the deviation between the predicted value and the true value to the ratio of the predicted value is its magnitude. When measured realistically, the total number of anticipations obtained is limited, and the optimal merit is used to replace the true value, and since the RMSE is very sensitive to the reflection of a larger or smaller error in a random set, the RMSE reflects the accuracy of the prediction very well. The formula for calculating the RMSE is:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\bar{y}_i - y_i)^2} \quad (21)$$

where: n denotes the total number of predicted samples; $\bar{y}_i$ denotes the predicted merit of the remaining life of the engine; and y_i denotes the true merit of the remaining life of the engine.

Score, scoring functions are often used to score performance in prediction problems to measure performance metrics such as accuracy, precision, and recall of the model. Also since RMSE has the same penalty for judging the comparison of the length of the predicted life and the true life, scoring functions

can compensate for the shortcomings of RMSE, i.e. later predictions impose a greater penalty on the scoring function. The scoring function is formulated as follows:

$$S = \begin{cases} \sum_{j=1}^m e^{-\left(\frac{l}{13}\right)} - 1, l < 0 \\ \sum_{j=1}^m e^{\left(\frac{l}{10}\right)} - 1, l \geq 0 \end{cases} \quad (22)$$

$$l = \text{RUL}_{\text{estimated}} - \text{RUL}_{\text{real}} \quad (23)$$

where: S is the final score; m is the number of engines in the test set; and l is the difference between true and predicted RUL.

4.3 Experimental analysis

In this chapter, the turbofan engine data published by NASA is used to verify the proposed method, and the data subsets of FD001 and FD003 are taken as examples to verify the network parameters of the PCLD model, and the influence of window size and step size on the PCLD model is analysed by univariate analysis. The proposed model is compared with DBRNN, LSTM-FNN, CNN and CNN-LSTM models. We compare the prediction results for series-parallel structures, and the prediction results for single-attentional and dual-attentional mechanisms.

Effect of different parameters: As shown in figures 9(a)–(d) are the comparison of prediction results for test set FD001 and

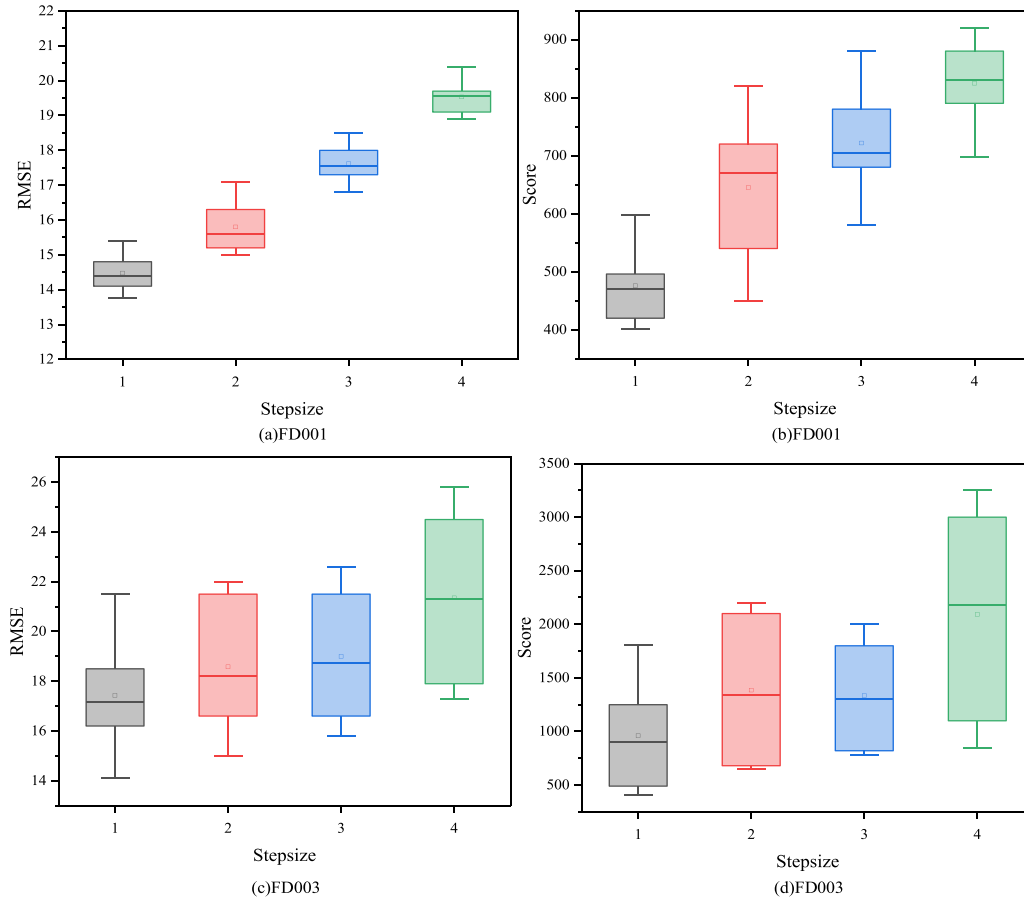


Figure 10. Predicted results at different step sizes.

test set FD003 under the window size of (5, 10, 15, 20, 25, 30), respectively. Figures 10(a)–(d) shows the comparison of prediction results for test set FD001 and test set FD003 with step size of (1, 2, 3, 4), respectively.

Since the size of the sequence with the shortest duty cycle in the FD001 and FD003 test sets is 31, the maximum window during parameter tuning is 30. As the window gradually increases, the amount of information contained in the data becomes larger, which is consistent with the actual situation. The effect of step size (1, 2, 3, 4) on the network is reflected in the prediction results figure 10. According to the result graph, it is seen that with the increase of the moving step size, the scoring function is getting bigger and bigger and the error is getting bigger and bigger, which leads to lower accuracy of the final prediction due to the loss of information from the model as well; through the comparison experiments of setting four step sizes, it is found that when the step size is set to 1, the loss function and the error are the smallest, and the final prediction result is the best. It can be seen that the nonlinear fitting ability can be enhanced with the increase of the number of units, which helps to improve the RUL prediction performance to some extent. However, when the number of units is

too large, the modelling ability of the model is limited due to the large amount of information, which leads to a decrease in performance.

The influence of different structures: We select engine data from either of the FD001 and FD003 data subsets to compare the series and parallel CNN and LSTM structures. As shown in figure 11, we found that the prediction accuracy of the series structure was lower than that of the parallel structure, and it was difficult to meet the practical prediction application. Therefore, the CNN-LSTM series structure works better in predictive model models. At the same time, as shown in figure 12, we compared the predictions under the application of DAM, temporal attention mechanism (TAM), characteristic attention mechanism, and no attention mechanism. Through comparison, it is found that the prediction accuracy of the model with an attention mechanism is higher than that of the model without an attention mechanism. The model of DAM is more accurate than that of applying one attention mechanism. In conclusion, the model with DAM has the best prediction accuracy. At the same time, we compared the prediction results before and after smoothing, as shown in figure 13, which is the comparison results of any engine data in FD001 dataset

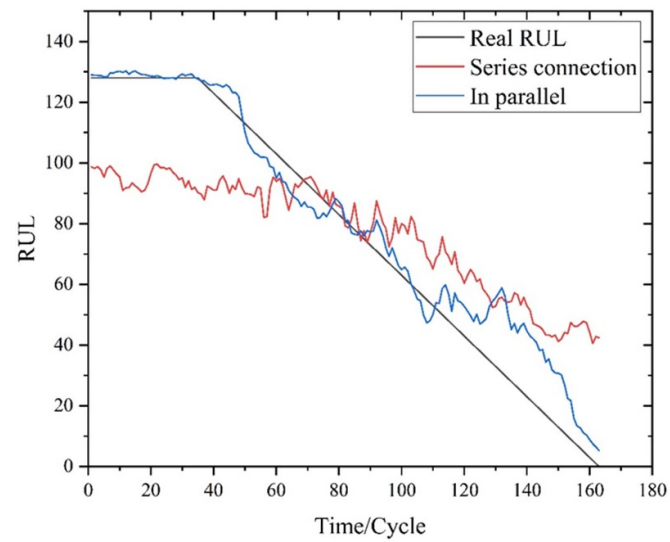


Figure 11. Prediction results of series parallel structure.

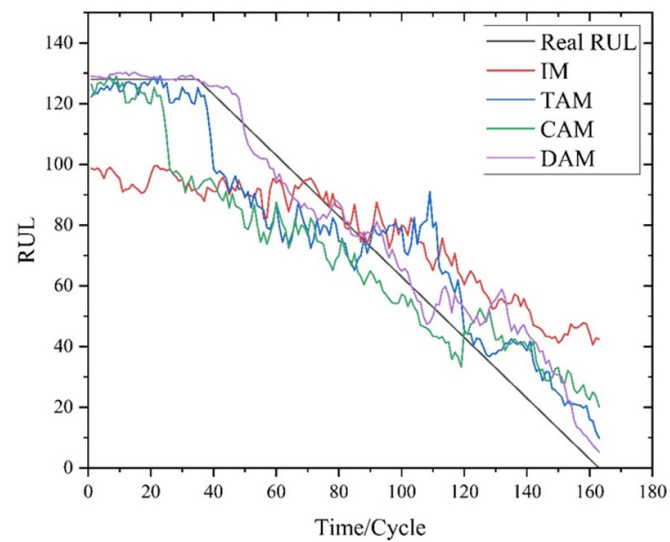


Figure 12. The predictive results of applying different attention mechanisms.

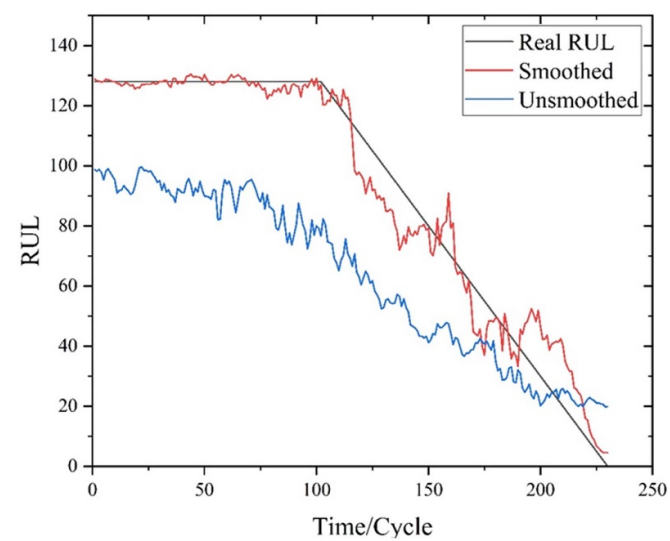


Figure 13. Comparison of prediction results before and after smoothing.

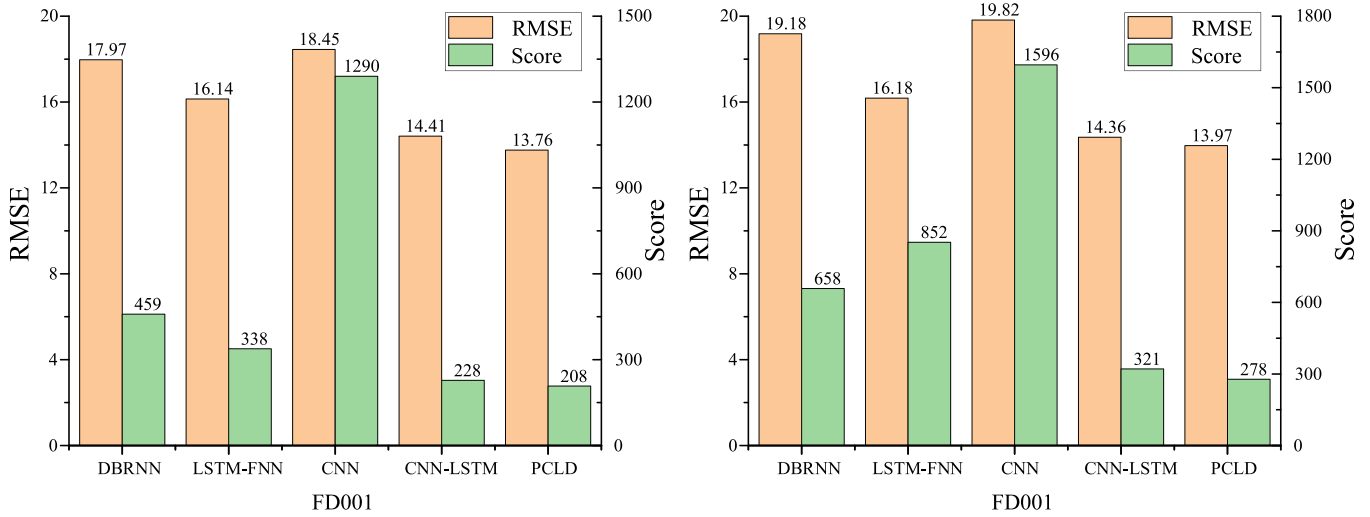


Figure 14. Predicted results of different methods.

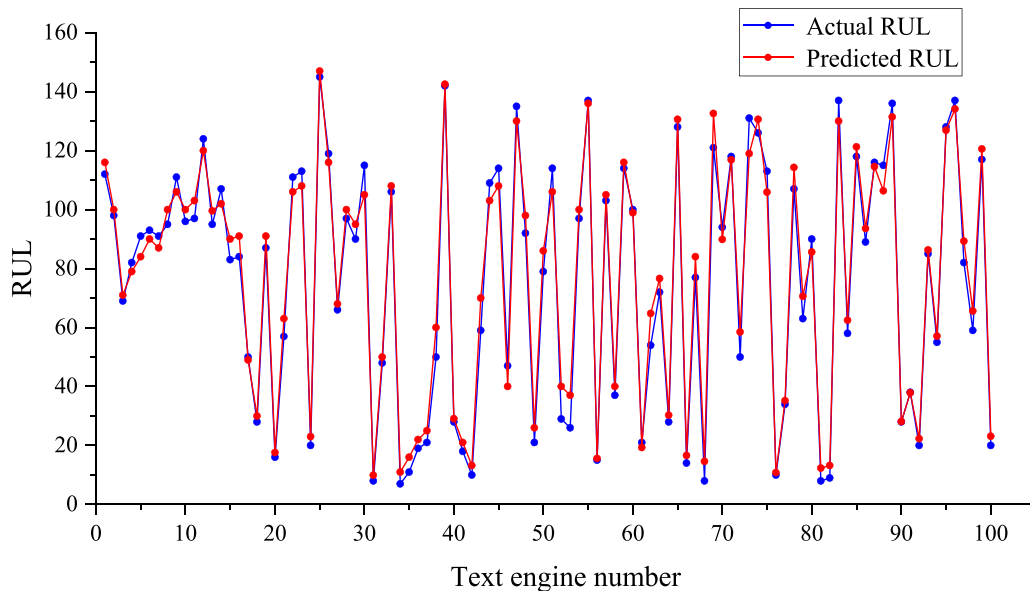


Figure 15. Prediction results for a subset of FD001.

before and after smoothing, and found that the accuracy of prediction was greatly improved after smoothing.

Comparison of methods: Comparison of other forecasting methods, and it is found through the result comparison plots that the PCLD model outperforms the predicted outcomes of other methodologies in terms of RMSE and Score on both data subsets. We found that the RMSE and scoring function of PCLD were the smallest compared with other models, indicating that the evaluation model had the smallest difference between the predicted value and the actual value, the best performance and the best prediction effect. The single neural network model CNN has the largest error of root mean square and the worst effect. The CNN-FNN results are better than the CNN and DBRNN results, but still lower than

the PCLD results, indicating that the dual model is better than the single model. The comparison between CNN-LSTM and PCLD fully demonstrates the feasibility and superiority of DAM. In summary, the proposed method has better performance and generalization ability than other methods. Figure 14 shows a graph comparing the prediction results of the various methods.

Analysis of the results: Changes in the architecture of deep learning methods and the use of attention mechanisms. The performance scores obtained from the predictions show that the new method has a higher prediction accuracy. So as to present the prediction results of PCLD clearly, comparing the predicted RUL with the actual RUL for all test units, as shown in figures 15 and 16 for the model PCLD on the

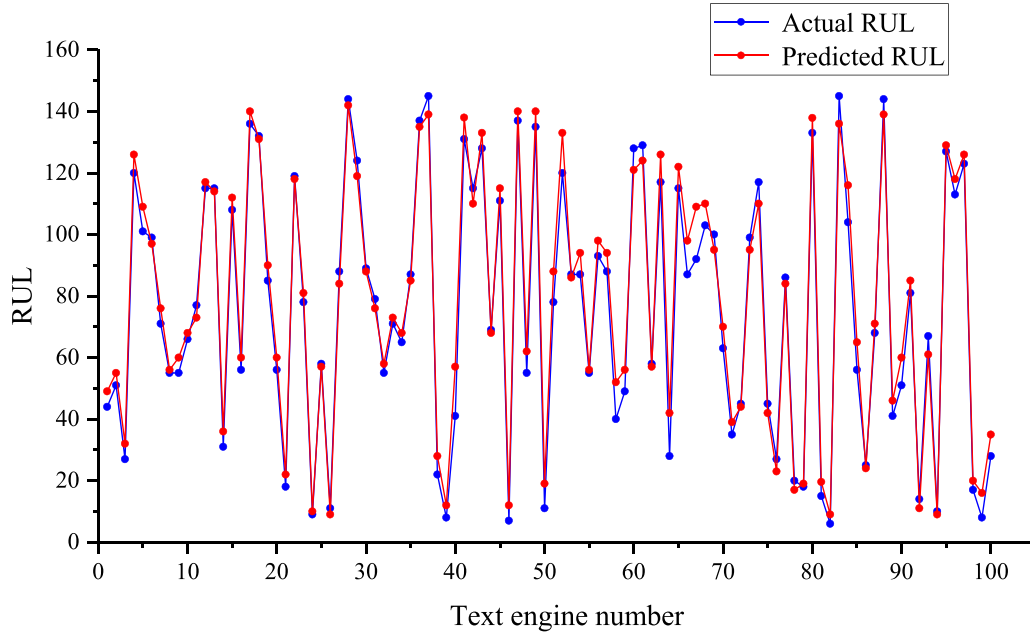


Figure 16. Prediction results for a subset of FD003.

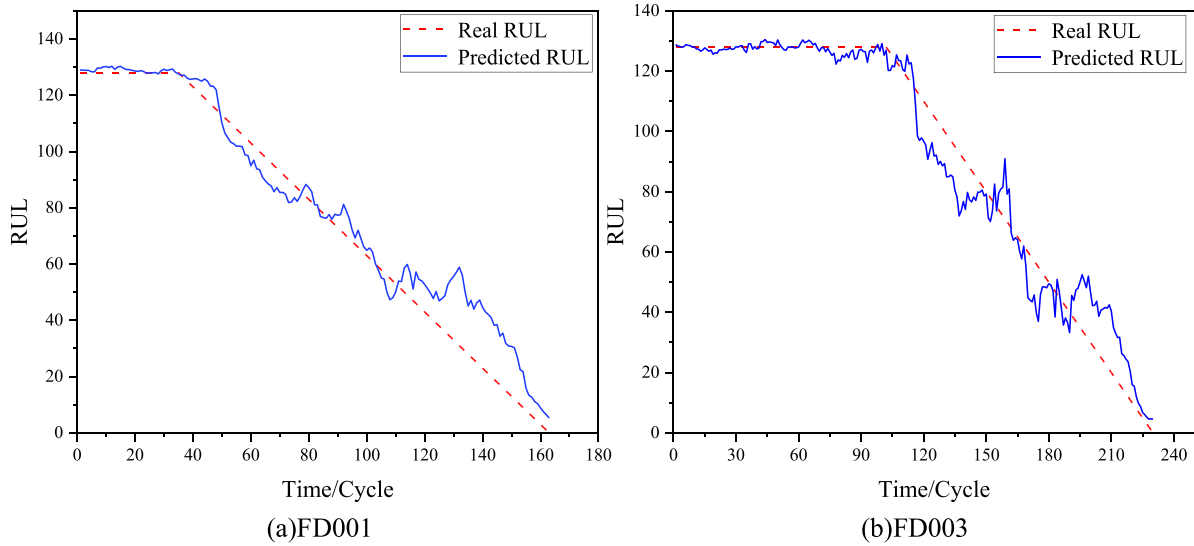


Figure 17. Prediction results from the stochastic engine.

data subsets FD001 and FD003, respectively. By analysing figures 15 and 16, the predicted RUL of all engine units are in general very close to the actual RUL, except for some engine units whose predicted RUL deviate from the actual RUL. In addition, in order to facilitate the observation of the consistency the predicted and real values of the motor, one engine from each of the two subsets is randomly selected for prediction, and figures 17(a) and (b) shows the prediction results of one random engine from subset FD001 and subset FD003 on the proposed method, respectively. According to the prediction results, the similarity between the real life of the aero-engine and the predicted value is high, so the prediction method in this paper has a good prediction accuracy.

5. Conclusions

This paper proposes a dual-attention mechanism guided parallel CNN-LSTM for turbofan engine lifetime prediction, in which parallel CNN-LSTM is mainly for excavating deep-seated feature in the extra-degenerate information to ensure that the information is fully mined at both the feature and time step. The DAM is simultaneously imposed on the parallel channel to improve the attention to the important information. Using the C-MAPSS data for experiments, methodology in this paper improves the accuracy of the RUL anticipation of turbofan engines to a certain extent, and the merits of this means can be proved in comparison with other methods. The parallel structure adopted by the method can fully explore the

important degradation information in the sensor monitoring data, and effectively solve the problem of important information being hidden or gradient disappearing in the prediction process; in which the dual-attention mechanism makes use of the important influences of time step and feature parameter in the prediction process, and assigning reweighting in the input element, so as to improve the identification of valuable information in the forecasting process. In this paper, the weighted moving smoothing process is also used to exclude the influence of complex noise prediction and highlight the life degradation trend, which is conducive to improving the prediction accuracy. Future research could incorporate transfer learning to improve the generalisation of the means with a small amount.

Data availability statement

The data cannot be made publicly available upon publication due to legal restrictions preventing unrestricted public distribution. The data that support the findings of this study are available upon reasonable request from the authors.

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Conflict of interest

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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